Synthesis of Butyl Enanthate via Fischer Esterification

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03/06/26

1 Introduction

The objective of this experiment is to synthesize butyl enanthate via Fischer Esterification by reacting 1-butanol with enanthic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

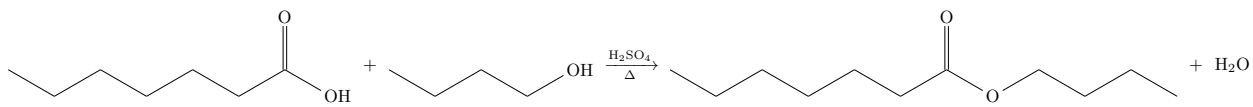


Figure 1: Reaction of Enanthic Acid and 1-Butanol

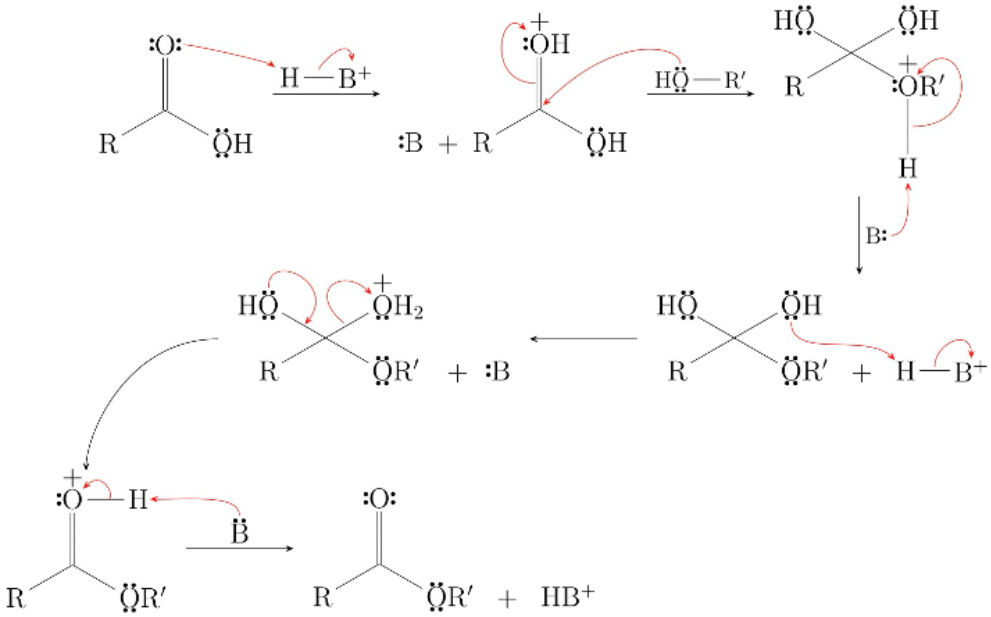


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Enanthic Acid	130.18	0.918	223	1.0	5.982 g / 6.52 mL	45.9
1-Butanol	74.12	0.81	117.7	3.0	10.217 g / 12.61 mL	137.8
Sulfuric Acid	98.08	1.84	–	cat.	5 drops	–
Diethyl Ether	74.12	0.710	34.6	solv.	–	–
Butyl Enanthate	186.29	0.856	226	–	8.56 g / 10.00 mL	45.9

4 Experimental Procedure

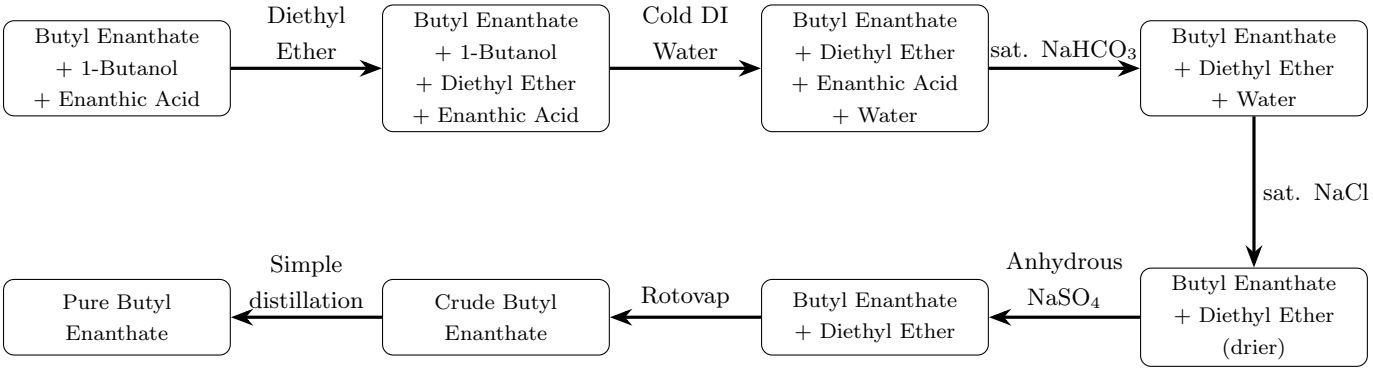
4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

4.2 Procedure

- In a 25 mL round-bottom flask, add 5.982 g (6.52 mL) of enanthic acid followed by 10.217 g (12.61 mL) of 1-butanol and a stir bar.
- While stirring, slowly add 5 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 20 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this three times.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude butyl enanthate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Enanthic Acid	g of 1-Butanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations